# Constructing trachea segmentation model based on deep learning

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# Content

Declaration ..... 1
Abstract ..... 4
Ⅰ Introduction ..... 5
1.1 Motivation ..... 5
1.2 Problem statement ..... 5
1.3 Approach ..... 6
1.3.1 Convolutional neural networks ..... 6
1.3.2 U-Net ..... 8
1.4 Project ..... 9
Ⅱ Technical Background ..... 11
2.1 Traditional methods ..... 11
2.2 Approaches Based on Machine Learning ..... 12
2.2.1 Machine Learning ..... 12
2.2.2 Deep Learning ..... 13
Ⅲ The Problem ..... 15
3.1 Traditional methods ..... 15
3.2 Deep learning methods ..... 15
Ⅳ The Solution ..... 17
4.1 Research Methods ..... 17
4.1.1 Data Sources ..... 17
4.1.2 Data Processing ..... 17
4.2 Model Design ..... 20
4.2.1 3D U-Net Model ..... 20
4.2.2 Deep Supervision ..... 21
4.3 Trachea segmentation ..... 22
4.3.1 Experimental Environment and Parameter Settings ..... 22
4.3.2 Adam Optimizer ..... 23
4.3.3 Evaluation Metrics ..... 23
Ⅴ Evaluation ..... 26
5.1 Results ..... 26
5.2 Explanation of results ..... 27
Ⅵ Conclusion ..... 29
References ..... 30
Appendix ..... 32

# Abstract

The segmentation of the trachea is an crucial aspect of medical image processing. CT image imaging frequently has problems with artefacts or partial volume effects. Traditional tracheal segmentation techniques exhibit low robustness and call for manual feature extraction. As a result, this study recommends a method that utilises deep learning for tracheal segmentation. This method employs U-Net as the fundamental architecture and integrates composite loss functions, with Dice, Acc, and Recall as the evaluation metrics. The study also confirms the impact of deep supervision strategy on experimental outcomes. The goal of this project is to automatically and precisely segment the tracheal area from CT images. A publicly accessible dataset is used to test the suggested methodology, and the results show how it can improve the precision and robustness of tracheal segmentation. Its broad potential for use in the field of tracheal segmentation is demonstrated by the results, which show strong performance, high segmentation accuracy, and robustness during testing.

**Key words**: Python; Trachea segmentation; Deep learning; U-Net; Convolutional neural network

# Ⅰ Introduction

## 1.1 Motivation

An essential organ for transferring gases between the human body and the outside environment is the trachea in the lungs. The trachea may become damaged over time through prolonged contact to the environment because of things like air pollution, chemical pollution, and radiation pollution. As a result, the prevalence of respiratory illnesses affecting the respiratory system of the lungs has increased. In 2018, there were 2.1 million new cases of lung cancer worldwide, making up the highest percentage of all new cancer cases at 11.6%, and 1.8 million deaths, making up the highest percentage of all cancer deaths at 18.4%, according to data from GLOBOCAN 2018 released by the International Agency for Research on Cancer (IARC) of the World Health Organisation[1]. Therefore, understanding the root causes and subsequent consequences of respiratory system-related illnesses is crucial for both the early detection and treatment of lung diseases.

The trachea of the lungs is a hollow tubular cavity composed of connective tissue, muscle tissue, and circular cartilage that interconnect to form it. About 17 levels divide the trachea from the main bronchus in the pharynx to the terminal bronchus, and as the levels increase, the tracheal radius and length decrease. The terminal trachea connects to the alveolar tissue, thereby completing gas exchange in the alveoli. There is a close association between lung diseases and tracheal lesions due to the trachea of the lungs' unique anatomical structure and physiological function. Changes in tracheal anatomy are often accompanied by tracheal lesions. As a result, the anatomy of the trachea in the lungs is frequently used to diagnose lung disorders and assess the effectiveness of treatment. And modifications to the tracheal anatomy frequently go hand in hand with lung tracheal lesions. As a result, physical details about the lung trachea are widely used in the diagnosis of lung disorders and the evaluation of the efficacy of treatment.

## 1.2 Problem statement

Due to their superior spatial resolution and imaging quality, computed tomography (CT) images are frequently used in the diagnosis of lung diseases. However, because of the numerous slices and intricate structure of the lung trachea tree, manual screening or segmentation of diseased tissue based on CT images is frequently labour- and time-intensive. Additionally, over time, the subjective nature of manual screening may result in incorrect diagnoses. Given the current workforce of medical professionals, increasing the number of doctors to conduct more thorough image analysis and organ segmentation tasks is not realistic. Because of this, automatic identification and pathological analysis of the lung trachea tree based on CT images can significantly lessen clinical workload, offer quantitative reference for clinical diagnosis, and lower the rate of misdiagnosis. In order to help

doctors read and annotate images, provide them with guidance, and effectively improve their work efficiency, reduce their workload, and ensure and improve the efficiency and quality of reading, achieving more detailed organ segmentation, an automated annotation method is required. Quantitative analysis and automatic diagnosis, depend on the automatic segmentation and acquisition of the lung trachea tree, which also serves as the foundation for other technologies like virtual bronchoscopy and surgical simulation navigation. An important stage in the interpretation of chest CT images is the segmentation of the trachea, which is crucial for clinical diagnosis and treatment. It can assist radiologists and other medical professionals in the accurate identification and diagnosis of lung disorders such lung cancer, pulmonary fibrosis, and chronic obstructive pulmonary disease (COPD). As a result, precise trachea segmentation is crucial for clinical practise. Trachea segmentation, however, is a difficult process because of the complicated shape and unequal grey distribution of the trachea's surroundings.

Traditional methods for tracheal segmentation rely on manual selection of features and parameters, which is limited in effectiveness and requires a lot of manual involvement. Tasks involving tracheal segmentation have become increasingly automated and effective in recent years thanks to the advancement of deep learning technologies. Tracheal segmentation techniques, which are based on deep learning, may automatically extract picture characteristics using convolutional neural networks, eliminating the need for manual feature selection and parameter configuration. This considerably improves the precision and efficiency of tracheal segmentation operations. The strong feature extraction abilities of deep learning have been widely used in medical image analysis tasks, particularly in the area of medical image segmentation, where deep learning technology has seen great success in improving segmentation accuracy, reducing manual intervention, and other areas. However, there are still some shortcomings and room for development.

# 1.3 Approach

## 1.3.1 Convolutional neural networks

In order to achieve image segmentation, convolutional neural networks (CNNs) are used in deep learning in this article. In computer vision and image processing, CNN is a crucial component of deep learning. Convolutional, activation, and pooling layers make up a CNN in most cases. Each convolutional kernel in the convolutional layer, which extracts features from input data, is made up of a weight coefficient and a bias, much like a neuron in a feedforward neural network. Each neuron in the previous layer is connected to multiple other neurons, and the size of the convolutional kernel determines how much of the image each neuron can perceive. The required features are calculated by the convolutional kernel as it moves over the image or feature map. Convolutional networks are better suited for image or video processing tasks than fully connected networks because they have fewer parameters, a stronger capacity for generalisation, and a better ability to prevent overfitting. Through

loss feedback on the training set, the convolutional neural network can optimise weights and ultimately extract more representative image features to boost the model's generalisation capabilities and segmentation accuracy, producing better segmentation outcomes. The process of neural network segmentation will then be further explained.

Convolutional neural networks extract features using convolutional operations. Convolutional operations can be viewed as straightforward weighted calculations, where the kernel's size serves as the convolution's receptive field and its parameters serve as its weights. One convolutional operation can be finished by sliding and weighting the kernel over the input image. In convolutional neural networks, there are two types of pooling operations: average pooling and max pooling. Average pooling chooses the average value of the pooling area as the pooling result, while max pooling chooses the maximum value of the pooling area. The size of the image changes following a pooling operation, but the total number of features stays the same. Convolutional neural networks employ pooling operations to efficiently lower the network's computational complexity, achieve multiscale feature extraction, and lower the possibility of overfitting the network.

The ultimate objective, regardless of whether it is a classification or segmentation task, is to classify using the extracted features. The difference is that the classification task produces a one-dimensional feature vector that records the category, while the segmentation task produces a two-dimensional feature map, which is the segmentation mask. Convolutional neural networks generally use fully connected layers to perform classification tasks. Therefore, the fully connected layer is usually located at the end of the network framework, used to transform the high-dimensional feature map into a low-dimensional feature vector. The essence of a convolutional neural network is to build a learning model using the basic operations mentioned above. Since convolutional neural networks have self-learning ability, they can implement end-to-end segmentation methods, which means that the segmentation results can be automatically generated after inputting the image without any additional operations. Similar to the human brain, deep convolutional neural networks can identify abstract features in images. The neural network can extract more intricate abstract features as the number of convolutional layers increases, greatly enhancing the network's generalizability.

Convolutional neural network-based artificial intelligence techniques have been used in a variety of fields recently thanks to improvements in hardware. In the field of clinical practise for medical image processing, a lot of AI-assisted diagnoses based on artificial intelligence have been used. The segmentation of medical images is a necessary step in computer-aided diagnosis, and the precision of the segmentation directly affects the diagnosis made by the doctor. In the last few years, the preferred technique for processing medical images has been supervised convolutional neural networks. They are frequently used in related tasks such as organ segmentation for computer-assisted diagnosis and lesion detection.

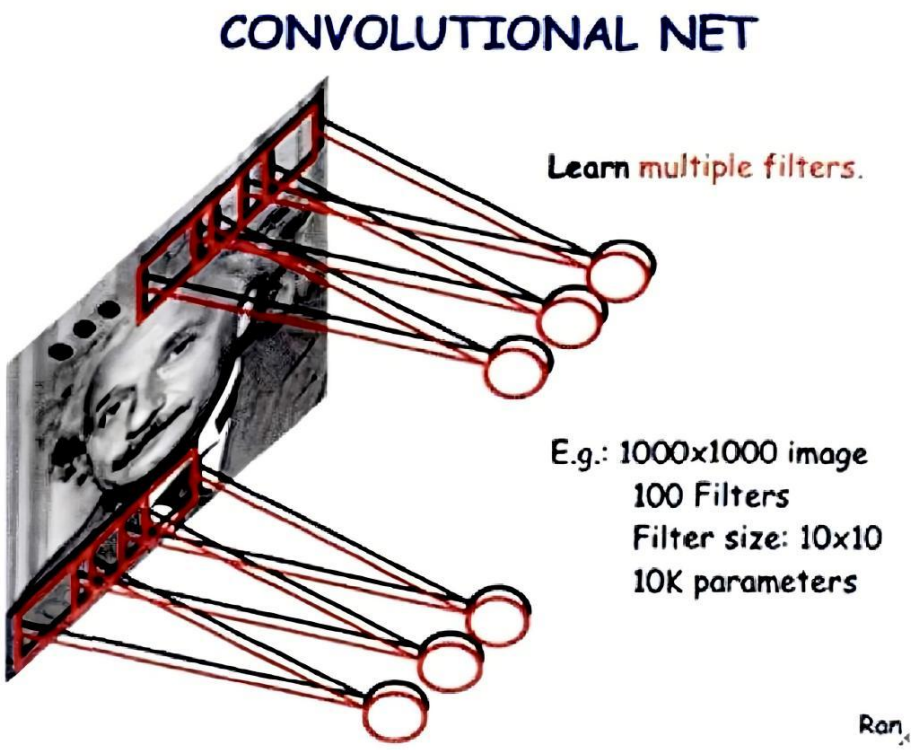



Figure 1-1 Convolutional Neural Network

### 1.3.2 U-Net

U-Net is a convolutional neural network (CNN) model used for image segmentation. The U-Net model, which can classify input images at the pixel level, is based on the fully convolutional network (FCN) architecture. The overall U-shaped structure of the U-Net network, which consists of an encoder, a decoder, and numerous skip connections, is depicted in Figure 1-2 U-Net Network Structure. With two 3x3 convolutional kernels in each layer of the encoder and decoder, U-Net performs feature extraction similarly to FCN by using multiple convolutional layers. A ReLU activation function is added after each convolutional layer to increase the nonlinearity of the system. The encoder, which has four downsampling processes, is specifically located on the left side of the U-Net. The size of the feature map is reduced using a 2x2 pooling layer with a stride of 2, ensuring that it is cut in half and that the number of channels is doubled after each downsampling. Shallow features are extracted through convolutional layers prior to each downsampling. The input image has now been compressed and its features have been extracted. The U-Net's decoder, which has four upsampling processes, is located on its right side. In order to gradually restore the original size of the input image, a 2x2 transposed convolution is used, which doubles the size of the feature map and reduces the number of channels. Prior to each upsampling in the decoder, the corresponding features from the downsampling layer are additionally combined through skip connections, which fuse the high-level semantic features extracted by the decoder with the low-level semantic features extracted by the encoder, preserving more semantic information and enhancing the accuracy of image segmentation.

The U-Net model has been extensively used in deep learning-based trachea segmentation projects and has produced noteworthy results. Its use and function are primarily seen in the following areas:

1. Effective feature extraction from images: The U-Net model has the ability to downscale and upscale at multiple levels, which allows it to extract high-quality features without sacrificing image details.
2. Use of the skip-connection feature: To preserve more of the image's fine details, the U-Net model can combine low-level and high-level features.
3. Lower computational cost: Because the U-Net model has fewer convolutional layers and parameters, it has a lower computational cost and can be trained and used quickly in environments with limited resources.
4. Higher segmentation accuracy: The U-Net model can segment medical images with greater robustness and segmentation accuracy. It can successfully identify the trachea and precisely segment the trachea contour.

The U-Net network structure is essential to deep learning-based trachea segmentation projects because of its strong robustness, straightforward structure, and high segmentation accuracy. Through its application, doctors' workloads can be reduced, segmentation accuracy and efficiency can be increased, and doctors can better understand patients' conditions to create more effective treatment plans.

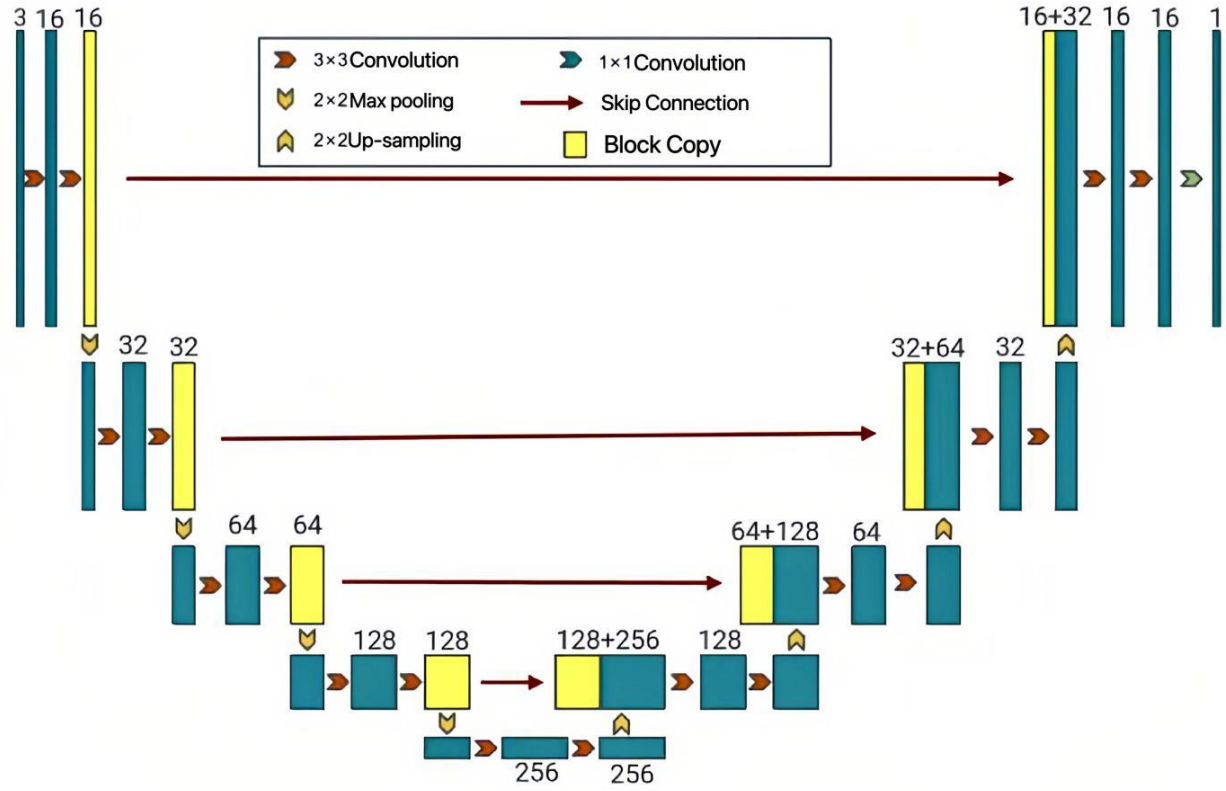



Figure 1-2 U-Net Network Architecture

## 1.4 Project

In order to increase the accuracy and effectiveness of trachea segmentation, this article suggests a deep learning-based method, using the evaluation metrics Dice, Acc, and Recall. The trachea in CT images is segmented in the experiment using a U-Net network, and the deep supervision strategy is used to increase the precision of the segmentation outcomes. The experiments also verified the effect

of deep supervision strategy on the experimental results.

The experimental results of this article show that the proposed deep learning-based trachea segmentation method achieves good segmentation performance on a public dataset, with higher accuracy and efficiency compared to traditional trachea segmentation methods, and has important clinical application value.

# Ⅱ Technical Background

The methods for trachea segmentation can be roughly divided into two categories: one is traditional methods, such as morphology-based methods, region-growing methods, and rule- and fuzzy logic-based methods; the other is detection methods based on tubular structures, which mainly rely on computing the Hessian matrix to identify the probability that points in the image are within the bronchial tree. These are the common mainstream methods for lung airway tree segmentation. The second category is machine learning-based methods, which use classification methods such as support vector machine (SVM) and k-nearest neighbor (kNN) to calculate the probability that a point is within the lung airway tree based on computed Hessian matrices, original CT values, and other information.

## 2.1 Traditional methods

By establishing specific rules for region growth and beginning from seed points that are either automatically or manually determined, the region-growing method is used in traditional methods to segment the lung airway tree. The segmentation effect of the algorithm is thus determined by the choice of seed points and the setting of growth rules. The front propagation method[2], which regulates the segmentation process by detecting the propagation state of the wavefront and judging and preventing leakage based on the size of the wavefront radius, is another approach comparable to 3D region-growing[3]. The tracheal wall and lumen's structure and grayscale characteristics are primarily used in morphology-based methods. In other words, the brighter tracheal wall surrounds the darker tube cavity in the two-dimensional slices of the CT scan, creating a local extremum. Therefore, to identify potential tracheal areas, morphology-based methods can identify the local extremum region. In most cases, 3D region growing is combined with morphology-based methods. The main bronchus is typically obtained using 3D region growing, and the small bronchi are then marked using morphology-based techniques. Finally, the subsequent fine segmentation is guided by the marked results. While morphology-based methods can identify more tracheal branches than single 3D region-growing methods, they do not take advantage of the trachea's 3D tubular structure. As a result, in situations where there is a lot of noise, it is simple to mark numerous false tracheal regions, which can cause local leakage when small bronchi are being extracted. The main structural features of the tracheal tree and prior knowledge of the tracheal tree and surrounding tissues are used to create classification rules or fuzzy logic in rule-based and fuzzy logic-based lung airway tree segmentation methods.

Despite being enhanced and combined, the aforementioned traditional methods for segmenting the trachea are still susceptible to noise and poor image quality. This led to the development of trachea segmentation techniques based on the detection of tubular structures in later research. The process of detecting objects in an image that resemble tubular structures using tubular structure filters is known

as tubular structure detection. Due to the trachea's inherent tubular anatomical structure, its features can be used to identify tubular structures in an image. The detected results can then be finely segmented to achieve trachea segmentation of the lung. Cheng et al.[4] analyzed the three-dimensional local features of pixel points using a multi-scale Hessian matrix to calculate the probability of a pixel point belonging to a tubular structure. The calculated results were then used to construct growth conditions, and finally the tracheobronchial tree was finely segmented using the three-dimensional region growing method. The tubular structure features defined by Cheng et al. are as follows:

$$\begin{cases} 0 \bullet \lambda_2 \bullet \lambda_3 \\ 0 \bullet \lambda_1 \bullet \delta^2 \end{cases} \tag{2-1}$$

In the equation, $\lambda_1$, $\lambda_2$ and $\lambda_3$ are the three eigenvalues of the Hessian matrix.

The method based on the analysis of tubular structures primarily makes use of the trachea's local and geometric features, making it more applicable than earlier methods and able to segment the trachea based on the segmentation case's actual tubular structure. It is more adaptable, less prone to leakage, and can further increase the number of detected tracheal branches than conventional methods. However, segmentation based solely on geometric features can easily result in breaks between tracheal branches and this type of method may result in missed detection of small airways in cases with low image resolution and high noise.

## 2.2 Approaches Based on Machine Learning

### 2.2.1 Machine Learning

In recent years, research has concentrated on the lung trachea tree segmentation method based on machine learning. When using machine learning techniques for trachea segmentation or leakage elimination, researchers primarily use the trachea's structure and grayscale features. The approach suggested in the literature by Lo et al[5]. is a good example. The segmentation results of leakage and non-leakage are first obtained using three-dimensional region growing, and a K-nearest neighbour (KNN) classifier is then trained using the two results[6]. In the segmentation cases, the axial directions of the trachea tree and the vascular tree are extracted concurrently using multi-scale Hessian matrix analysis. The final trachea tree is segmented based on the local direction difference between the trachea and the blood vessels and the classification outcomes of the classifier. The machine learning-based approach introduces the ideas of learning and training, which enables the algorithm to carry out autonomous learning and accurately classify the tracheal and non-tracheal regions, effectively eradicating segmentation leaks. There is currently no literature or reports on the effects of feature selection and optimisation issues on tracheal segmentation results, despite the fact that the features used to train the classifier and the choice of the training set can have a significant impact on the classification results. The number of detected tracheal branches and the length of the trachea still need to be improved, despite the segmentation results obtained using machine learning methods having the lowest false positive rate.

### 2.2.2 Deep Learning

Deep learning is a crucial part of machine learning and a technique for learning features from data. It can mimic the neural architecture of the human brain and comprehend data by mimicking how the brain processes information. It benefits from strong robustness, high automation, and high accuracy. Utilising deep learning techniques for lung trachea segmentation has the benefit of a trained model with strong generalisation capabilities[7] and the ability to complete large-scale segmentation quickly[8]. Deep learning techniques for tracheal segmentation have the additional benefits of effectively suppressing missegmentation and minimising segmentation leakage in addition to the advantages already mentioned. As a result, numerous studies on the segmentation of blood vessels and bronchi by neural networks have been conducted recently. Deep learning addresses the issue of traditional methods' limited capacity for generalisation. Deep learning techniques can complete lung image segmentation tasks of various qualities by removing abstract features from images. In addition, current deep learning-based segmentation methods primarily use end-to-end technology, which automatically calculates the input data through a network model and outputs the segmentation result, significantly reducing the need for manual interaction, streamlining the organ segmentation process, and shortening the segmentation time.

In the field of medical image object detection, Zhu et al. [13] proposed a Faster RCNN object detection framework based on 3D convolution for the detection of pulmonary nodules. They also used an encoding-decoding structure similar to Unet for benign-malignant classification of nodules. In terms of segmentation, some optimization has been done on the region-growing method. For example, Duan et al. used a 3D region-growing method to obtain the main bronchus and then obtained candidate regions in three directions based on the bronchial plane features in each direction. They optimized the segmentation results in each direction using the candidate regions and finally merged them to obtain the final segmentation result[14]. Some researchers have used hybrid methods. For instance, Bauer et al. used gray-level, local shape, and feature structure to segment images and optimized the segmentation results using graph structures[15], while Qier et al. used a combined filtering method to preprocess images and used SVM to identify points belonging to the bronchial inner points among the candidate points[16]. To enhance the bronchus, an enhancement filter was applied based on the characteristic that the gray value difference between the bronchus cavity and bronchial wall is large. Lassen et al.[19] proposed using the 9 different directions of each pixel as 2D planes and enhancing small bronchi by determining whether the pixel is surrounded by the bronchial wall in each plane. Harrison et al.[12] developed a method based on 2D convolution and used 2D P-HNN to extract features from each slice layer. For 3D medical images such as CT and MRI data, methods based on 2D convolutional neural networks usually apply convolution to single-layer slices to extract 3D segmentation results. For example, Zhou et al.[9] applied a 2D fully convolutional neural network to 3D image data to extract the pancreas. Qi et al.[11] developed a 3D deep supervision network to segment the liver from abdominal CT images. Juarez et al. [20] proposed a deep learning-based 3D

Unet algorithm to extract the trachea.In addition to offering for the first time 20 sets of training data and 20 sets of test data for trachea segmentation, the EXACT09[21] trachea segmentation competition offered a platform for assessing trachea segmentation accuracy. The 20 test data sets include different imaging parameters and imaging devices in addition to covering common lung diseases seen in clinical practise. The EXACT09 competition also put forth a comprehensive set of evaluation metrics, such as the total number of tracheal branches, the total length of the trachea, the leak volume, and the false positive rate. Numerous techniques were developed during the competition, and both the techniques and the evaluation metrics have had a big influence on subsequent research. Additionally, it has recently been discovered that segmentation techniques based on 3D CNN that make use of 3D spatial data and integrate context can produce better segmentation outcomes. One such example is the V-Net, which was initially put forth by Milletari et al.[10] for the segmentation of the pancreas but has since been used to segmentation of other parts due to its excellent segmentation outcomes. In comparison to conventional methods, deep learning has significantly improved performance overall. Modern clinical settings frequently employ deep learning-based computer-aided diagnosis techniques, which have contributed to significant advancements in the processing of medical images.

The two techniques become more complex, each with unique advantages. Both methods have seen corresponding research advancements in recent years. The number of branch points and the length of the trachea in the segmentation results have been greatly improved, and the false positive rate has also been greatly reduced, according to a comparison of domestic and foreign pulmonary tracheal segmentation methods over a long period of time. Additionally, the segmentation results evaluation indicators have been improved. Finally, it was discovered that limiting segmentation leakage and achieving higher tracheal detection rates are frequently incompatible after summarising and analysing the research. Strict segmentation guidelines can prevent leakage, but they also eliminate potential tracheal regions. In contrast, the likelihood of leakage rises. Therefore, a major obstacle impeding the advancement of pulmonary tracheal tree segmentation technology is how to further enhance the tracheal tree's segmentation effect while guaranteeing a low false positive rate.

# Ⅲ The Problem

## 3.1 Traditional methods

Based on the aforementioned analysis, it can be concluded that the traditional segmentation idea takes into account both tracheal segmentation and leakage detection during the segmentation process, that is, detecting and preventing leakage during the tracheal segmentation process. The number of detected tracheal branches is reduced as a result of this segmentation method's frequent rejection of numerous potential tracheal regions. As a result, the development of lung tracheal tree segmentation technology is currently constrained by the need to further enhance the tracheal tree's segmentation effect while maintaining a low false positive rate. Similar to blood vessels, the trachea also has a tubular shape and coexists with them. Its wall also has a grayscale value. Additionally, due to the tracheal wall's poor visibility in small tracheas and the tracheal lumen's similarity to lung tissue's grayscale value, lung tissue can easily be missegmented, resulting in leakage. Traditional methods for segmenting the lung airways are cumbersome and have a low degree of automation, and there are certain drawbacks.

## 3.2 Deep learning methods

While there are many benefits to using deep learning methods to segment the lung airways, there are also a number of drawbacks. First, there is a dearth of annotated data for training networks in lung airway segmentation tasks, which means there are many unlabeled data that cannot be used for training and few labelled medical image data. So, in addition to studying network structures and loss function optimisation, researchers have taken a data-centric approach to their work in order to collect as much training data as they can. For segmentation optimisation, researchers like Zhao et al. are dedicated to learning more prior bronchial information[17]. In order to increase the amount of trainable data for deep learning, Jin et al. have studied how to use network simulation of real data. However, the fidelity still needs to be improved. Additionally, Irving et al.[18] proposed a technique to extract the airway's branches in various directions using 3D grayscale reconstruction; however, it is comparatively ineffective and also has issues with missing small branches. Second, there is little research on how to create medical image datasets. Since most medical images are three-dimensional, the segmentation accuracy will vary depending on how the dataset is created. Finally, in terms of segmentation precision and network model size, the U-Net network still has space for development. The network structure must be improved to better match the characteristics of lung medical images, the number of network parameters must be decreased, and segmentation accuracy must be increased. Additionally, due to the size of medical images, block learning is required in order to segment the images using deep convolutional networks, which inevitably results in the loss of some global information.

Deep learning methods can be divided into two types: 2D and 3D. In order to perform semantic

segmentation, the 2D learning method first extracts features from depth maps and images, then merges them. Convolutional neural networks (CNN) are frequently used in these techniques to extract features. Some of these techniques increase segmentation accuracy by cascading multiple features. The benefit of this approach is that it is appropriate for 2D image segmentation tasks, has a lower computational cost, and has a faster training speed. The 3D learning method analyses and processes 3D images to finish segmentation. The 3D CNN or 3D convolution-based network architectures are required for these techniques. The computational complexity of 3D data is higher than that of 2D learning methods due to the complexity of the data, necessitating the use of more computing resources and a longer training period. The ability to extract spatial information from 3D data is improved by 3D models. This results in the widespread use of 3D learning techniques in 3D medical imaging, particularly in the segmentation of medical imaging data like CT and MRI. Therefore, 3D learning methods typically produce better results when segmenting 3D data, such as the trachea. In contrast, 2D learning techniques have the benefit of quick computation, but they are limited in their ability to extract airway branches from 3D data because they struggle to make good use of 3D spatial information. This method has some drawbacks because the three-dimensional representation of the airway tree requires spatial data in three dimensions. On the other hand, 3D learning techniques can extract 3D spatial data more effectively but use more computational power.

# Ⅳ The Solution

In this study, we proposed a deep learning-based method for trachea segmentation, mainly using a 3D U-Net network for accurate segmentation. This article elaborated on this trachea segmentation method from various aspects including research methods, model design, experimental settings, evaluation metrics, and model testing.

## 4.1 Research Methods

### 4.1.1 Data Sources

In the data processing, we used a publicly available dataset: 40 scans from EXACT '09. The EXACT '09 challenge provides 20 scans in the training set and 20 scans in the testing set. All CT scan axial dimensions are 512 × 512 pixels with a spatial resolution of 0.5~0.781 mm. The number of slices per scan ranges from 157 to 764. Their slice spacing is between 0.45-1.0 mm. We divided these CT images into training, validation, and testing sets, which make up 80%, 10%, and 10% of the total.

### 4.1.2 Data Processing

We performed a number of data preprocessing operations, including normalisation, resampling, lung window extraction, and threshold truncation, to prepare CT images for neural network training. The main goal of data preprocessing is to remove noise and redundant data from the image data, which enhances the model's performance and stability.

1. Normalization:

Min-Max Normalization: Min-Max Normalization is an important preprocessing method, the purpose of which is to eliminate the differences between CT scanning equipment and make the image data comparable and unified. In this experiment, we used a simple linear normalization method to scale the pixel values of the original CT images to the range of [0,1]. Specifically, we performed the following transformation on each pixel value:

$$pixel_{\text{norm}} = \frac{\text{pixel} - \min_{\text{pixel}}}{\max_{\text{pixel}} - \min_{\text{pixel}}} \tag{4-1}$$

where pixel represents the pixel value of the original CT image, $\min_{\text{pixel}}$ and $\max_{\text{pixel}}$ represent the minimum and maximum values of the pixel values, and $\text{pixel}_{\text{norm}}$ represents the normalized pixel value.

Z-score normalization: Z-score normalization is a commonly used normalization method based on the standard normal distribution, which maps each data point in the sample space to the corresponding point on the standard normal distribution. The formula for Z-score normalization is:

$$x_{norm} = \frac{x - \mu}{\sigma} \tag{4-2}$$

where $\mu$ represents the mean value of the original data, $\sigma$ represents the standard deviation of the original data. $x_{norm}$ is the new value obtained after normalization, which usually includes negative values. Z-score normalization can map data to a standard normal distribution with a mean of 0 and a variance of 1, and therefore has good comparability and interpretability. It is widely used in machine learning, data mining, and other fields.

2. Resampling

The pixel spacing on the x, y, and z axes was standardized to 1mm. The purpose of this step is to reduce computational complexity and memory consumption, and to ensure that CT images obtained from different scanning devices have consistent pixel spacing. Different datasets may differ in size and resolution as a result of the variable size and resolution of CT imaging data. To eliminate these differences, we used trilinear interpolation to resample the original CT images to a fixed size of 512x512x256 in this experiment, and standardized the image resolution to 1x1x1 mm.

3. Lung Windowing:

Lung windowing is an image processing technique commonly used in medical imaging for accurate localization and reconstruction of lung regions. In medical imaging, the contrast and grayscale values of different tissues and organs may vary greatly, and many non-tracheal tissues and structures may make it difficult to clearly identify and recognize the lung region. To reduce interference with model training, we performed lung windowing on the CT image. Lung windowing adjusts the brightness and contrast of medical images to make the display of the lung region more clear and prominent, in order to facilitate better medical diagnosis and accurate localization.

Specifically, lung windowing is an image processing technique commonly used for precise localization and image reconstruction of the lung region in medical imaging. In medical images, there may be significant differences in contrast and grayscale values among different tissues and organs, and many tissues and structures unrelated to the trachea may make it difficult to identify and clarify the lung region. To reduce interference with model training, we performed lung windowing on the CT images. Lung windowing involves filtering and highlighting pixels with grayscale values in a certain range in medical images. Since the bronchus is only distributed in the area where the left and right lung tissues are located and occupies a small volume in the entire CT data, extracting the lung region and removing irrelevant backgrounds is very important for improving the segmentation effect of the bronchus. The preprocessing steps for lung region extraction are as follows: (1) Select a threshold based on medical experience, and the thresholding method can obtain a mask of the external muscle area with higher HU values (the grayscale value of CT voxel points). Use the flood fill algorithm to obtain the mask of the human body background outside the image. (2) Combine the muscle mask and human body background mask, binarize the image, and then invert it to obtain the mask of the lung tissue. (3) Perform a bitwise AND operation between the mask of the lung tissue and the original CT to obtain the region of interest in the lung.

In this project, the HU value range of the CT image during lung windowing is generally [-1024,1024], and the HU value of air is -1000. To better recognize the trachea features and eliminate irrelevant information, the HU value of the CT image is adjusted to [-1024,-700] through threshold truncation.

4. Thresholding:

The goal of thresholding is to turn an image into a binary image by setting all pixel values in the input image that are lower than a specific threshold to 0 and all pixel values that are higher than the threshold to 1 (or 255). Thresholding is frequently used in deep learning for image segmentation tasks like trachea segmentation.

Thresholding is crucial for segmenting the trachea and can significantly raise the accuracy of deep learning models' segmentation. The use of thresholding in trachea segmentation is frequently referred to as "gray-scale truncation". Since the trachea area in the image is brighter than the tissue around it, a suitable threshold can be chosen to identify the trachea area as pixels with values greater than the threshold and the tissue around it as pixels with values lower than the threshold. The performance of the model's prediction can be assessed using these markings, which can also be used to produce training and validation data.

Gray-scale truncation may be impacted in practical applications by elements like noise and resolution, so it is important to select the right threshold based on the circumstances. A few adaptive thresholding techniques are currently available that can choose the ideal threshold based on the distribution of grey values in an image. In conclusion, thresholding is a widely used image processing technique that, when applied to this project, can help us more accurately achieve trachea segmentation, laying the groundwork for further medical image analysis.

5. Sliding Window Sampling:

Sliding window sampling is a commonly used image processing technique, which is usually used for block processing or cropping of a specific part of an image. The principle of this technique is to move a sliding window of a fixed size over the image and pass each image block within the window as input to the model for feature extraction or prediction. By sliding the window, all possible regions of interest in the image are covered, improving the model's ability to extract target features. Overlapping can also be used to increase prediction accuracy. At the same time, sliding window sampling reduces the input size of the model, avoiding the problem of excessive computational complexity, and is one of the commonly used preprocessing techniques.

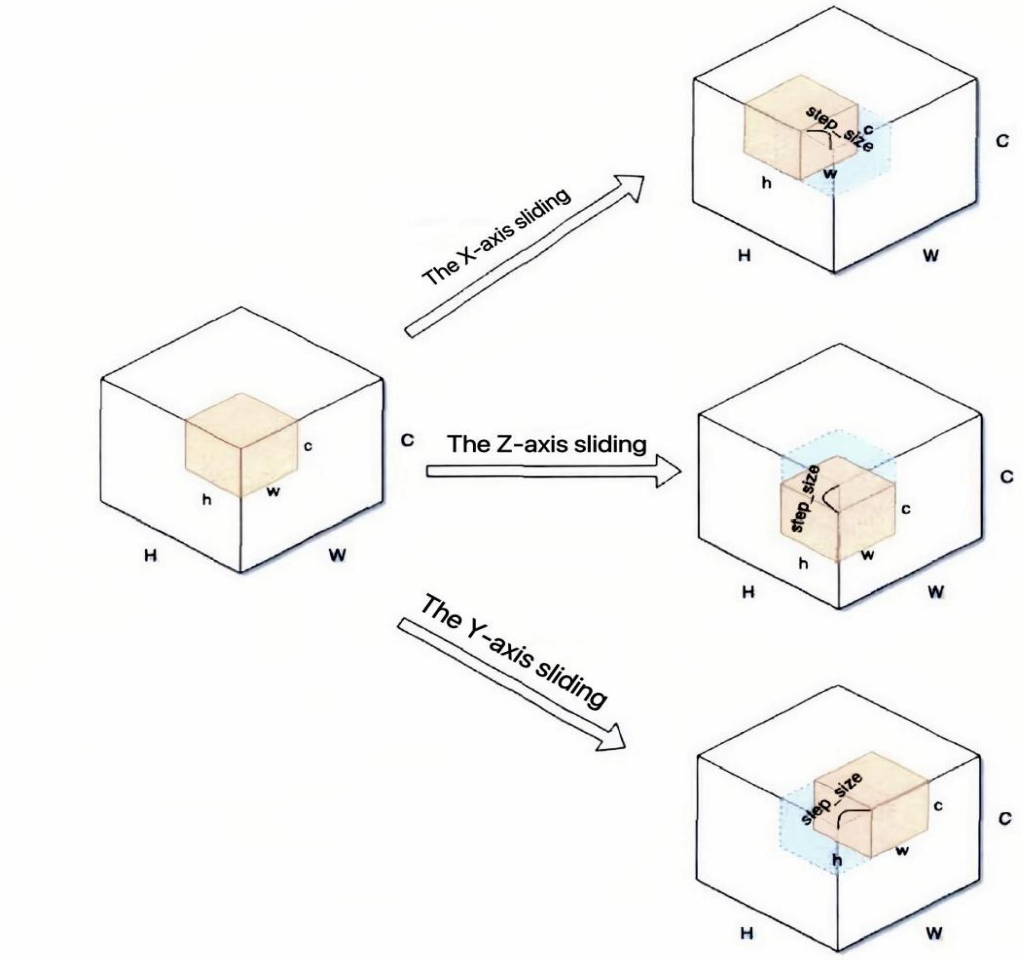



Figure 4-1 Sliding Window Sampling

After completing the five steps listed above, we were able to obtain preprocessed CT image data, which gives the trachea segmentation task that follows a more reliable and accurate starting point.

Finally, we save all medical images and their corresponding trachea segmentation labels to the local file system for model training and testing purposes.

# 4.2 Model Design

## 4.2.1 3D U-Net Model

In order to improve segmentation's precision and efficiency for the trachea application scenario, which has the characteristic of tubular structure, a suitable segmentation method must be selected. When compared to 2D approaches, 3D approaches are better suited for this situation. As a result, the trachea region can be located and segmented more precisely while maintaining the continuity of the image structure, further enhancing the stability and reliability of trachea segmentation. This is because 3D methods can fully consider the three-dimensional information in the trachea data. As a result, in this particular situation, 3D methods are more advantageous and a more logical and scientific choice. The 3D U-Net model's

application in this project is crucial. The trachea region can be precisely extracted from trachea images using this model, offering a crucial foundation for medical professionals' diagnosis. Additionally, the model can lessen the amount of manual intervention by doctors, increase segmentation accuracy and efficiency, and greatly enhance the accuracy and efficiency of medical work.

The model consists of multiple 3D convolutional layers, ReLU activation functions, and 3D pooling layers. In the last layer of the model, we used the sigmoid function to output binary results. U-Net is a classical fully convolutional neural network used for image segmentation tasks. It has an encoder-decoder structure, which can effectively extract image features and perform pixel-level classification. As an extension of U-Net, 3D U-Net can handle 3D images and has stronger representation and better segmentation performance than traditional 2D U-Net when processing 3D images. Figure 4-2 depicts the structure of the 3D U-Net. To extract image features, the encoder uses convolutional and pooling layers to gradually decrease the image resolution while increasing the number of feature maps. The decoder has several layers of 3D transpose convolution and upsampling that can restore the encoder's output to an output image with the same dimensions as the input image. Skip connections are used in the corresponding encoding and decoding modules to preserve more spatial information, assisting the model in more accurately capturing features at various scales.

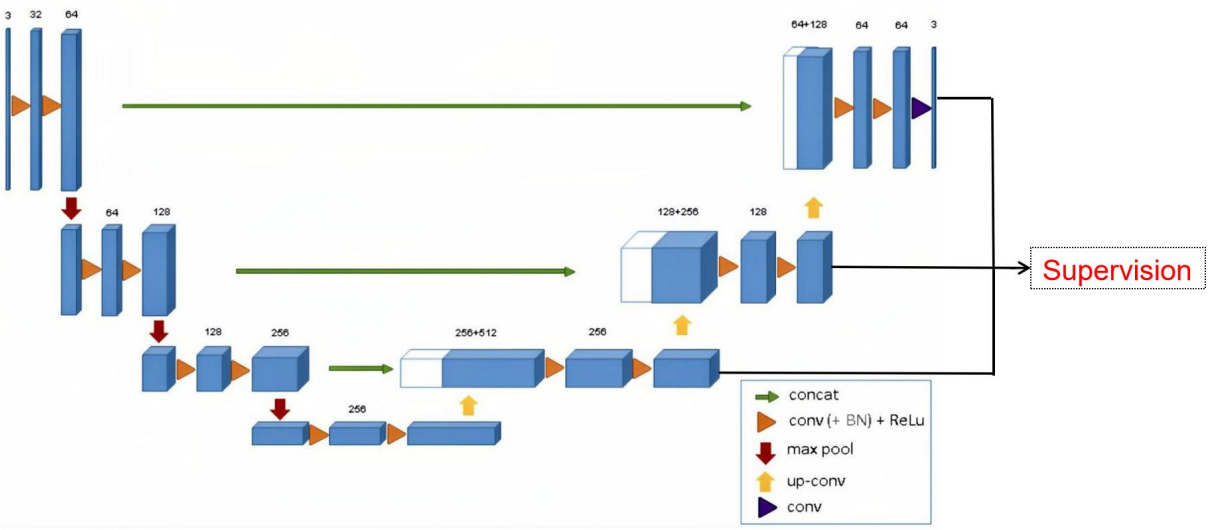



Figure 4-2 3D U-Net Architecture with Deep Supervision

## 4.2.2 Deep Supervision

Deep supervision is a multi-task learning technique used in deep learning that can enhance a network's segmentation performance and training effectiveness. To obtain feature information at various scales, it extends the network in multiple directions. The outputs of these branches are then combined to produce more precise results. This method has the advantage of allowing the network to gradually accumulate more fine-grained feature data during the training process, increasing the precision of the segmentation outcomes.

We used the deep supervision method in this project to enhance our segmentation model. In order to calculate the difference between the feature maps at various scales and the true labels, we added

numerous branches to the network, each with a different loss function. We can get more precise segmentation results by combining the differences between feature maps at various scales and the true labels.

In particular, we extended our model by two branches, one of which produces a feature map at a scale of 1/2 and the other at a scale of 1/4. Each branch computes the difference between its feature map and the true labels using a composite loss function. The network parameters are then optimised using backpropagation after adding the loss functions of the two branches.

## 4.3 Trachea segmentation

### 4.3.1 Experimental Environment and Parameter Settings

The experiments in this chapter were conducted on a device in the laboratory with the following specifications: CPU, 32GB RAM, and NVIDIA 3070Ti GPU. The algorithm was implemented mainly using the Python3 language and the open-source Pytorch framework environment. The specific experimental environment is shown in Table 4-1.

Table 4- 1 Experimental Device and Environment Configuration

| Name | Configuration |
|---|---|
| Device | CPU：Intel®Core™i9-12900H RAM：32GB Windows11 GPU：3070Ti |
| Programming language | Python 3.9.16 |
| Deep learning framework | Pytorch 1.9.0 |

Regarding training, a total of 20 samples were used in this chapter's experiments, and the Adam optimizer was used for gradient descent. The best model was saved after training. The main parameter settings are shown in Table 4-2.

Table 4- 2Experimental Main Parameter Settings

| Parameter | Set value |
|---|---|
| Batch_size | 2 |
| Patch_size | 128×128×32 |
| Learning rate | $1\times10^{-3}$ |
| Optimizer | Adam |

### 4.3.2 Adam Optimizer

An essential part of deep learning that is utilised to train neural networks is the optimizer. To reduce the loss function as much as possible will help the neural network better suit the data. Based on the gradient of the loss function, the optimizer modifies the neural network's parameters in order to continuously enhance prediction accuracy. Using various optimizers can significantly affect the training outcomes in deep learning.

In this project, we used an optimizer based on stochastic gradient descent called Adam to accelerate model training and convergence. Adam is an adaptive learning rate optimization algorithm that combines momentum and adaptive learning rate methods. The momentum method speeds up model training, while the adaptive learning rate method effectively avoids the problem of the learning rate being too large or too small.

The Adam algorithm's fundamental concept is to estimate the gradient's first- and second-order values and then use those values to update each parameter's learning rate. Specifically, the Adam algorithm maintains two state variables, one for the first-order estimate (the average gradient) and the other for the second-order estimate (the average square of the gradient). It then uses these state variables to calculate the learning rate for each parameter.

In our trachea segmentation project, the use of the Adam optimizer makes model training more stable and allows it to quickly find the minimum of the loss function. Through experimental comparison, we found that the Adam optimizer can achieve better training results in a shorter period of time, improving the accuracy of trachea segmentation.

### 4.3.3 Evaluation Metrics

In deep learning, a loss function is a function used to evaluate the difference between the predicted results of a model and the true labels. Its role is to guide the updating of model parameters to make the model's predicted results closer to the true results. In this project, we used a composite loss function as the model's loss function.

The Dice coefficient is a segmentation model measurement index based on the overlapping area, and its calculation formula is as follows:

$$\mathrm{Dice}(T,P)=\frac{2*(T\cap P)}{|T|+|P|} \tag{4-3}$$

The symbol |•| represents the number of voxels, T represents the ground truth label, and P represents the predicted label. The calculated Dice value is between 0 and 1, with values closer to 1 indicating better segmentation performance.

Cross-entropy loss is a common classification loss function used for multi-class problems. In this project, we need to convert the trachea segmentation problem into a binary classification problem, i.e.,

to divide the pixels in the image into two classes: trachea and non-trachea. Therefore, we used a binary cross-entropy loss function.

The calculation formula for binary cross-entropy loss function is:

$$L_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log\left(p_i\right) + \left(1-y_i\right)\log\left(1-p_i\right)\right] \tag{4-4}$$

where N represents the number of samples, $y_i$ is the true label of the i-th sample, and $p_i$ is the probability that the model predicts the sample as the trachea. This loss function can directly calculate the cross-entropy between the model's predicted results and the true labels, thereby evaluating the performance of the model.

During model training, we use this loss function as the model's objective function and update the model parameters through the backpropagation algorithm to minimize the loss function. In each iteration, the model performs forward propagation calculation based on the training data, obtains the cross-entropy loss between the model's predicted results and the true labels, then uses the backpropagation algorithm to calculate the gradient of the loss function with respect to the model parameters, and finally updates the model parameters through an optimizer.

The Dice coefficient and cross-entropy loss function are two commonly used evaluation metrics and loss functions in deep learning, each of which has pros and cons of its own. The Dice coefficient is widely used in binary image segmentation tasks and is suitable for evaluating the segmentation effect of algorithms on small targets or low-contrast images, especially in the analysis of medical images with relatively complex segmentation targets. The calculation method of the Dice coefficient is simple and can well consider the similarity between the predicted and true segmentation results, especially in dealing with class imbalance. However, the Dice coefficient does not take into account the relationship between true segmentation result pixels and background pixels, so it cannot locate the background pixels of the model well. In addition, the Dice coefficient has little impact on pixels with low prediction confidence, making its performance weak in dealing with some complex situations.

The cross-entropy loss function can effectively mine the feature information of data and has good performance in dealing with the imbalance of positive and negative samples. The cross-entropy loss function, however, has trouble handling situations where the predicted results contain correlated information, such as target segmentation, where pixels from various parts of the same target may be correlated. The cross-entropy loss function has a limited capacity to handle this complex information crossover. Additionally, the cross-entropy loss function's performance may be affected by cases with significant noise interference because it is sensitive to variations from the predicted results.

Accuracy is also a commonly used metric in machine learning to measure the performance of a binary classification model. It measures the percentage of correct predictions made by the model out of all predictions made.

The formula for accuracy is:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{4-5}$$

where:

true positives (TP) are the number of correctly predicted positive samples

false positives (FP) are the number of negative samples that were incorrectly predicted as positive

true negatives (TN) are the number of correctly predicted negative samples

false negatives (FN) are the number of positive samples that were incorrectly predicted as negative

Recall is a performance metric used to evaluate the ability of a binary classification model to correctly identify positive instances from the total number of actual positive instances. It is defined as the ratio of true positive predictions to the total number of actual positive instances, i.e.,

$$Recall = \frac{TP}{TP + FN} \tag{4-6}$$

where TP is the number of true positive predictions and FN is the number of false negative predictions.

Recall is a measure of completeness because it captures how well a model is able to capture all positive instances, even at the cost of making more false positive predictions. It is particularly useful in scenarios where identifying all positive instances is more imprtant than minimizing false positives.

# Ⅴ Evaluation

## 5.1 Results

Test results are shown in the tables：

Table 5-1 Dice

| | Case16 | Case17 | Case18 | Case19 | Case20 |
|---|---|---|---|---|---|
| Deep Supervision | 89.15% | 94.31% | 93.70% | 83.85% | 66.06% |
| without Deep Supervision | 87.28% | 93.71% | 91.51% | 83.84% | 84.22% |

Table 5-2 Acc

| | Case16 | Case17 | Case18 | Case19 | Case20 |
|---|---|---|---|---|---|
| Deep Supervision | 99.97% | 99.98% | 99.99% | 99.94% | 99.91% |
| without Deep Supervision | 99.97% | 99.98% | 99.98% | 99.94% | 99.91% |

Table 5-3 Recall

| | Case16 | Case17 | Case18 | Case19 | Case20 |
|---|---|---|---|---|---|
| Deep Supervision | 90.09% | 97.25% | 95.42% | 79.59% | 85.11% |
| without Deep Supervision | 86.29% | 97.83% | 95.97% | 79.59% | 85.11% |

The experiment uses ITK-SNAP software to generate visualization results and compare them with the label model, using case17 as an example, as shown in Figure 5-1, where a is the label, b is the predicted result, c is part of the details of the label, and d is the same part of the details of the predicted result.

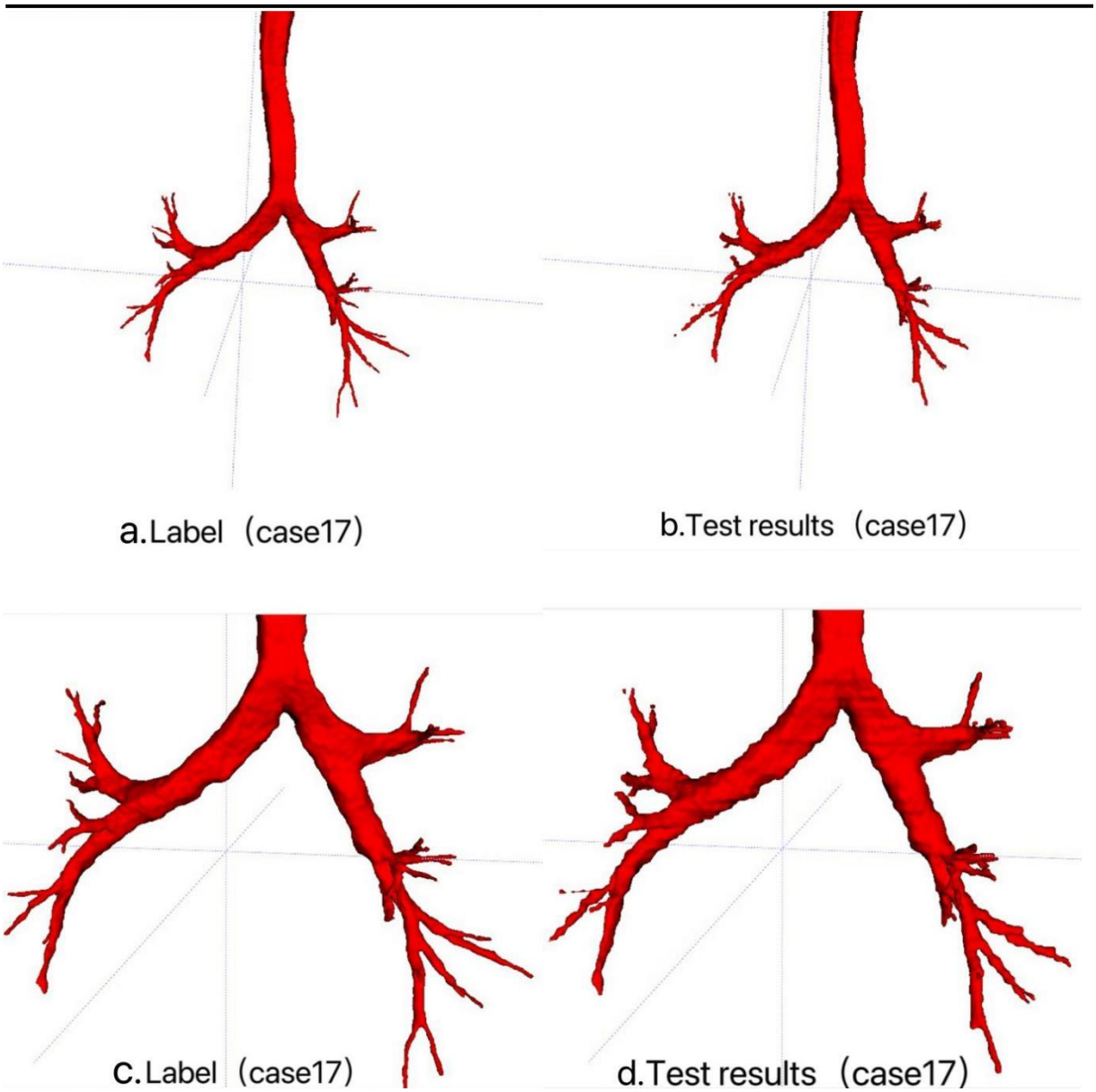



Figure 5-1 Visualization results of case17

## 5.2 Explanation of results

The test results show that our model performs well on the test dataset. With deep supervision, the average Dice coefficient is 0.8541, the average Acc coefficient is 0.9996, and the average Recall coefficient is 0.8949. This indicates that our model can effectively segment the trachea in CT images, achieving the expected segmentation performance with high accuracy and stability.

However, due to label leakage in Case20, the reference value of this test data is limited. After removing this test case, the average Dice coefficient with deep supervision is 0.9025, the average Acc coefficient is 0.9997, and the average Recall coefficient is 0.9058. Without deep supervision, the average Dice coefficient is 0.8909, the average Acc coefficient is 0.9997, and the average Recall coefficient is 0.8992. Comparing the experimental results, it can be seen that with deep supervision,

the average Dice coefficient is 1.16% higher than without deep supervision, indicating that the model can have better performance and obtain more excellent trachea segmentation results with deep supervision.

In addition, we further analyzed the model and found that the segmentation performance of the tracheal edge and some detail parts inside the trachea still needs to be improved. We believe this may be due to three reasons: (1) insufficient training data; (2) low image quality, resulting in some blurry images; (3) inconsistent data sources. Therefore, we plan to further expand the training data, improve the image standard, unify the data source, and explore more effective trachea segmentation methods in future work.

# Ⅵ Conclusion

In this study, we propose a deep learning-based method for trachea segmentation, which utilizes the U-Net network as the basic architecture and combines a composite loss function with Dice, Acc, and Recall as evaluation metrics. We also validate the effect of deep supervision on the experimental results. The results show that our model achieved a Dice coefficient of 90.25%, an accuracy of 99.97%, and a recall of 90.58% on the test set, demonstrating its high efficiency and accuracy, and its potential to improve accuracy and efficiency in clinical applications. In the future, we will continue to explore better deep learning models to further improve trachea segmentation performance. Additionally, we will investigate how to apply this method to other medical image segmentation tasks. Currently, this method is only suitable for segmenting the trachea and needs to be further extended to other organs and lesions in the lung, such as lung parenchyma, pulmonary vessels, and fissures.

In summary, the deep learning-based trachea segmentation method proposed in this paper is of great significance in the field of medical image analysis. Through a series of experiments, we have improved the robustness and generalization ability of the model, achieving satisfactory results. Our approach overcomes the disadvantages of traditional methods, such as tedious segmentation steps and low automation level for trachea segmentation in the lung. In the future, we will continue to optimize the model and apply it to a wider range of medical image analysis to improve clinical diagnosis and treatment accuracy and efficiency.

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# Appendix







## Maynooth International Engineering College

### RECORD OF MEETINGS

**Record of Meetings Form:** All students are required to download a copy of this form and bring it to every project meeting. You must complete the form and have it signed by the supervisor. In the case of a virtual meeting, the form should be completed and signed by your supervisor at a later stage. You are required to submit a digital copy of all meeting forms in the appendix of your final report.

### Final Year Project 2022 – 2023

| Meeting Date | Student Signature | Supervisor Signature |
|---|---|---|
| 2023.03.06 | SHEN HONG | Panlin |
| 2023.03.13 | SHEN HONG | Panlin |
| 2023.03.20 | SHEN HONG | Panlin |
| 2023.04.01 | SHEN HONG | Panlin |
| 2023.04.05 | SHEN HONG | Panlin |
| 2023.04.07 | SHEN HONG | Panlin |
| 2023.04.12 | SHEN HONG | Panlin |
| 2023.04.15 | SHEN HONG | Panlin |
| 2023.04.20 | SHEN HONG | Panlin |
| 2023.04.27 | SHEN HONG | Panlin |